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Chapter 30. Manuscript frames 1359–1384. Catalogue snapshot: 5 October 2026.

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Chapter 30

GELFAND-P001359

23.3.87 Gelfand seminar.

V. E. Zakharov. $\bar{\partial}$-problems and integrable systems.

Construction of integrable equations by the inverse-problem method. (actually neither integrable nor by the inverse-problem method)

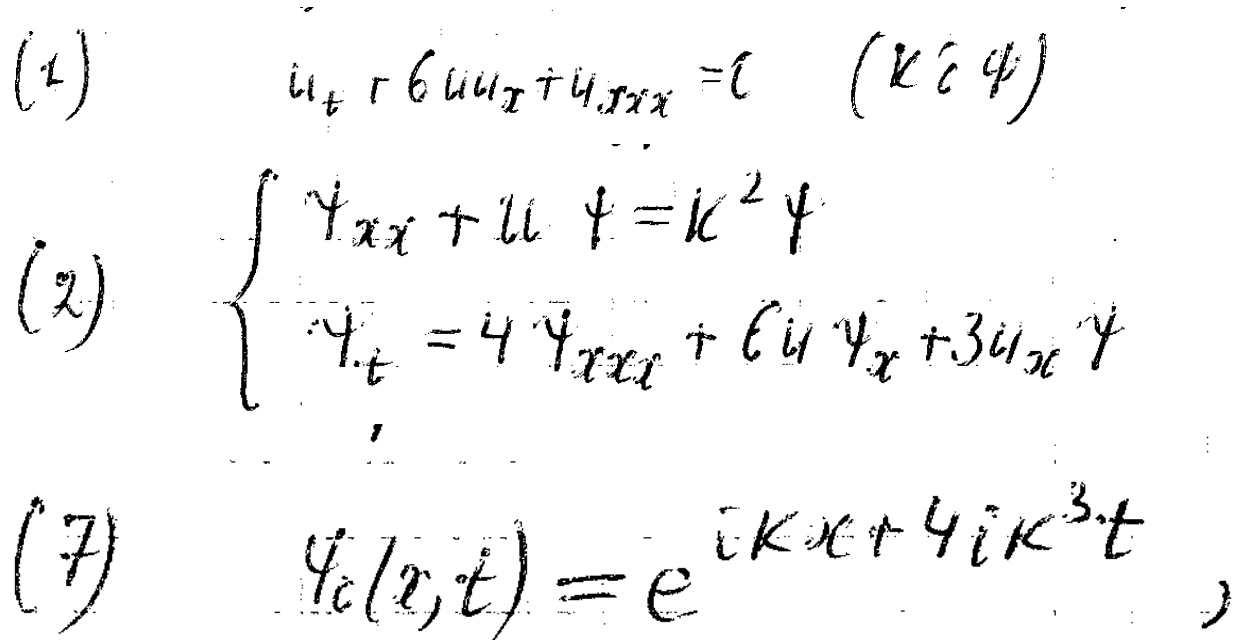
1973 A. B. Shabat formulated the dressing method.

Nonlinear equations are considered as compatibility conditions for overdetermined linear systems

$$u_t + 6uu_x + u_{xxx} = 0 \qquad \text{(KdV)} \tag{1}$$

— the compatibility condition for the equations

$$\begin{cases} \psi_{xx} + u\psi = k^2\psi, \\ \psi_t = 4\psi_{xxx} + 6u\psi_x + 3u_x\psi. \end{cases} \tag{2}$$

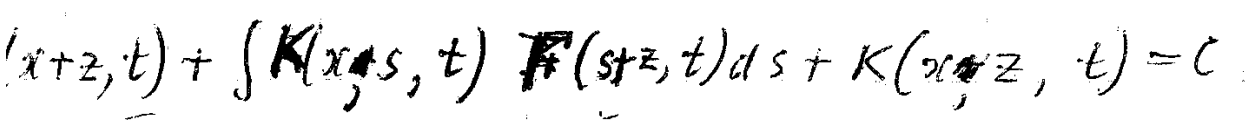


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Uncertain reading. The sign before $6uu_x$ is incomplete and is read as $+$. Compare the signs in (2) and (7).

The method consists in considering the equation

$$F(x+z,t) + \int K(x,s,t)F(s+z,t)\,ds + K(x,z,t) = 0. \tag{3}$$



Original source details: GELFAND-P001359

Uncertain reading. In the arguments of K, plus signs are crossed out and commas placed below. The letters underneath the retraced K and F are illegible.

for any function F for which

$$\frac{\partial}{\partial t}F(\xi,t)=8\frac{\partial^3 F}{\partial \xi^3}. \tag{4}$$

In (3), the integral is understood in one of three ways:

$$\int=\int_x^\infty \quad \text{or} \quad \int_{-\infty}^x \quad \text{or} \quad \frac{1}{2}\left(\int_x^\infty \ldots - \int_{-\infty}^x \ldots\right). \tag{5}$$

Then

$$u=2\frac{d}{dx}K(x,x,t) \quad \text{— a solution of (1).} \tag{6}$$

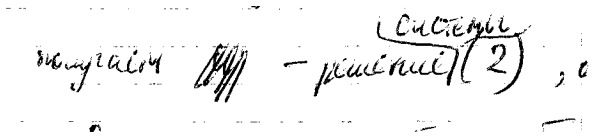
Taking

$$\psi_0(x,t)=e^{ikx+4ik^3\cdot t}, \tag{7}$$

and then

$$\psi=\psi_0(x)+\int K(x,s)\psi_0(s)\,ds, \tag{8}$$

we obtain [crossed out: illegible symbol, possibly ψ?] — a solution of [inserted above the line: the system] (2), from which u is found.



Uncertain reading. illegible symbol, possibly ψ

Original source details: GELFAND-P001359

Consider also more general functions F satisfying the equations

$$\begin{cases}\dfrac{\partial^2 F}{\partial x^2}-\dfrac{\partial^2 F}{\partial z^2}=0,\\ \dfrac{\partial F}{\partial t}=4\left(\dfrac{\partial^3 F}{\partial x^3}+\dfrac{\partial^3 F}{\partial z^3}\right).\end{cases} \tag{9}$$

GELFAND-P001360

[source number: 24]

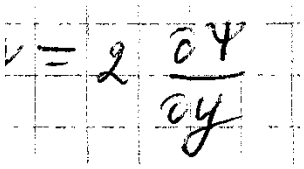
Then everything will hold as before.

Generalization. In place of $(9)_1$ we write:

$$\frac{\partial^2 F}{\partial x^2}-\frac{\partial^2 F}{\partial z^2}=\alpha\frac{\partial F}{\partial y}. \tag{10}$$

Then

$$\begin{cases}\psi_{xx}+u\psi=2\dfrac{\partial \psi}{\partial y},\\ \psi_t=4\psi_{xxx}+6u\psi_x+3u_x\psi\end{cases} \tag{11}$$



Uncertain reading. The coefficient in (11) reads 2. The α in (10) and (12) has a different shape.

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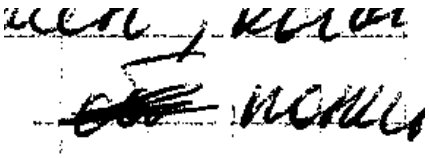
and for u we obtain:

$$\frac{\partial}{\partial x}\left[u_t + 6uu_x + u_{xxx}\right] = 3\alpha^2 u_{yy}. \tag{12}$$

y becomes a new parameter in (3).

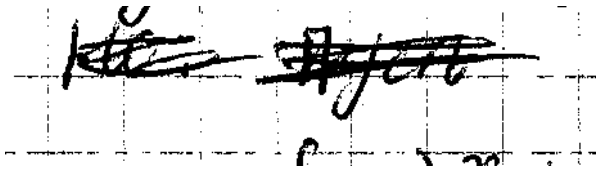
This was in 1974 in the work of Zakharov and Shabat.

The third integral in (5) is defined when the function decays in both directions. However, there is another way to [crossed out: illegible beginning of a word?] interpret the integral:



Uncertain reading. illegible beginning of a word

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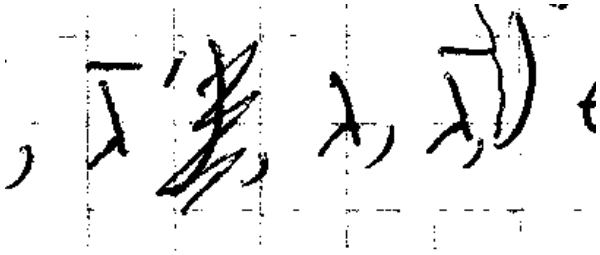
Uncertain reading. Crossed out on the line below: possibly “as Fourier”.

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$$\int e^{\lambda x}\,dx = \frac{1}{\lambda}e^{\lambda x} \qquad \text{and so on} \tag{13}$$

We will interpret the integral in the third way in (5). Introduce the function T by the formula

$$F(x,z) = \frac{1}{2\pi}\iint T(\lambda',\bar\lambda',\lambda,\bar\lambda,t,y)\, e^{i(\lambda' x - \lambda z)}\,d\lambda'\,d\bar\lambda'\,d\lambda\,d\bar\lambda. \tag{14}$$



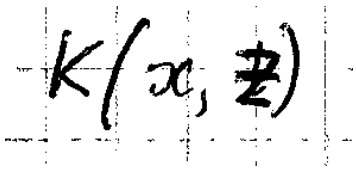
Uncertain reading. An illegible symbol after $\bar\lambda'$ is crossed out.

Original source details: GELFAND-P001360

[Inserted above the line before the closing parenthesis: t, y.]

and define the function K as

$$K(x,z) = -\frac{1}{2\pi}\int K(\lambda,\bar\lambda,x,t,y)e^{i\lambda(x-z)}\,d\lambda\,d\bar\lambda. \tag{15}$$



Uncertain reading. The second argument on the left is z, written over an earlier illegible symbol.

Original source details: GELFAND-P001360

[Inserted above the line before the closing parenthesis: t, y.]

This is a way to find growing solutions of the G.–L.–M. equation.

If K and F satisfy the integral equation, all is well (we use convention (13)).

[crossed out: $\chi(x, y)$.] Set:

$$\chi(\lambda, \bar{\lambda}, x, y, t) = 1 + \frac{1}{2\pi i} \int \frac{K(\eta, \bar{\eta}, x, y, t)}{\lambda - \eta}\, d\eta\, d\bar{\eta}. \tag{16}$$

Here:

$$\frac{1}{\lambda - \eta} = \lim_{\varepsilon \to 0} \frac{\bar{\lambda} - \bar{\eta}}{|\lambda - \eta|^2 + \varepsilon^2}. \tag{17}$$

Let us write the equation:

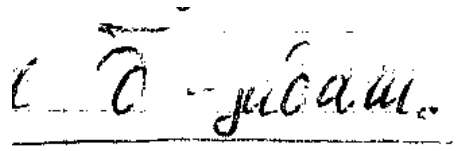
GELFAND-P001361

$$\frac{\partial \chi}{\partial \bar{\lambda}} = \int \chi(\eta, \bar{\eta}, x_i) T(\eta, \bar{\eta}, \lambda, \bar{\lambda}, x_i)\, d\eta\, d\bar{\eta}. \tag{18}$$

$$x_i = (x, y, t)$$

This is equivalent to the original approach.

The dressing method in the language of $\bar{\partial}$-problems.



Uncertain reading. The heading may end with "problems" or "problem".

Original source details: GELFAND-P001361

Let T satisfy the system.

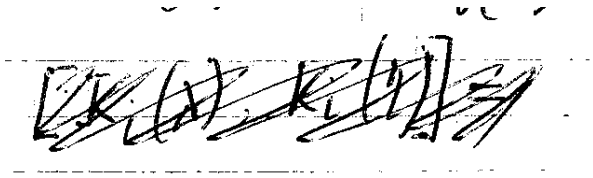
$$\begin{cases} \dfrac{\partial T}{\partial x} = -i(\lambda - \eta)T, \\ \dfrac{\partial T}{\partial y} = -\alpha(\lambda^2 - \eta^2)T, \\ \dfrac{\partial T}{\partial t} = -4(\lambda^3 - \eta^3)T, \end{cases} \tag{19}$$

from which

$$T = T_0(\eta, \bar{\eta}, \lambda, \bar{\lambda}) e^{-i(\lambda-\eta)x - \alpha(\lambda^2 - \eta^2)y - 4(\lambda^3 - \eta^3)t}. \tag{20}$$

In place of (19) one can introduce $x = x_i = (x_1, x_2, x_3)$ and the functions

$$K_i(\lambda) \quad \text{and} \quad I_i(\lambda)\,:$$



Uncertain reading. In (21) a line resembling $[K_i(\lambda), K_i(\eta)] =$ is crossed out. The subscripts and the end are uncertain.

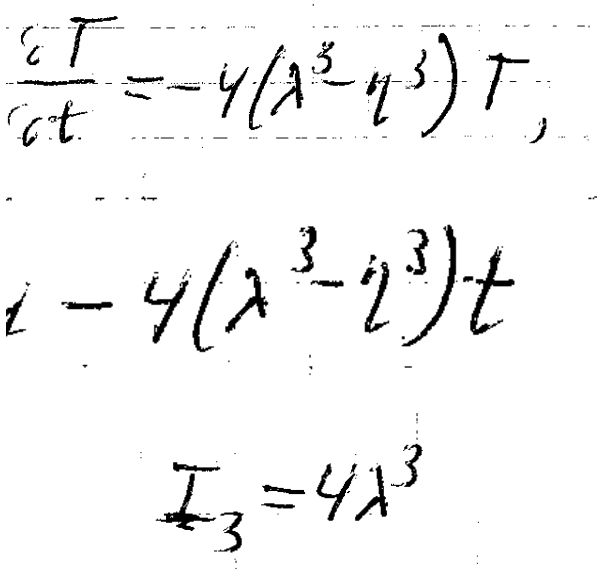
Original source details: GELFAND-P001361

$$[I_i(\lambda), I_j(\eta)] = 0. \tag{21}$$

$$\frac{\partial T}{\partial x_i} = I_i(\eta)T - TI_i(\lambda). \tag{22}$$

In the example

$$I_1 = i\lambda, \quad I_2 = \alpha\lambda^2, \quad I_3 = 4\lambda^3. \tag{23}$$



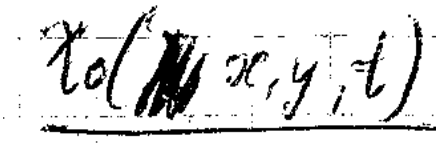
Uncertain reading. The cubic terms in (19), (20), and (23) have 4 without i.

Original source details: GELFAND-P001361

Now solve the nonlocal $\bar{\partial}$-problem (18) with the normalization condition $\chi(\infty) = 1$.

Let

$$\chi = 1 + \frac{\chi_0(x, y, t)}{\lambda} + \dots \tag{24}$$



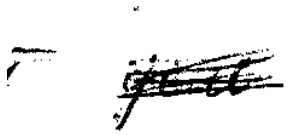
Uncertain reading. An argument before x, y, t in the numerator is crossed out, possibly λ.

Original source details: GELFAND-P001361

Then

$$u = 2i\frac{\partial \chi_0}{\partial x} \tag{25}$$

satisfies [crossed out: possibly "the formula"?] the KP equation, (12).



Uncertain reading. possibly "the formula"

Original source details: GELFAND-P001361

GELFAND-P001362

[source number: 25]

(26) Main assertion:

For K_j one can take any three commuting rational matrix functions.



Original source details: GELFAND-P001362

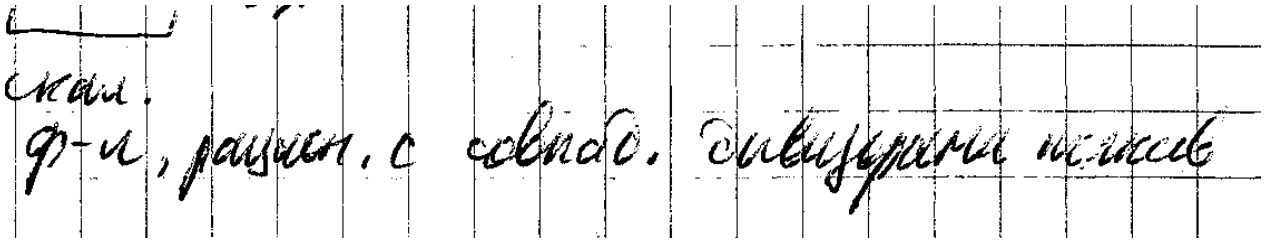
Uncertain reading. The K is written over an earlier illegible sign.

One can also take functions on a Riemann surface.

(27) New proposal.

take $K_i(\lambda)$ to be commuting differential operators with coefficients rational in λ

$$L_i = \frac{\partial}{\partial x_i} + K_i(\lambda) + \underbrace{f_i(\lambda)}_{\substack{\text{scalar functions, rational,} \\ \text{with freely moving poles}}} \frac{\partial}{\partial \lambda}. \tag{28}$$



Original source details: GELFAND-P001362

Uncertain reading. The final word of the caption is more likely “полюсов” than “полюсами” (both mean “poles” here).

(29) $[L_i, L_j] = 0 \Rightarrow$ conditions on K_i in which the number of equations $=$
$=$ the number of unknowns.

Many of the resulting equations are new (if everything has variable coefficients).

The same can be done on an elliptic curve.

In the work of Belinsky/Zakharov (1978), a special case is gravity.

New this year:

$$L_i = \frac{\partial}{\partial x_i} + M_i\left(\lambda, \frac{\partial}{\partial \lambda}\right). \tag{30}$$

$$[L_i, L_j] = 0 \quad \Longleftrightarrow \quad [M_i, M_j] = 0. \tag{31}$$

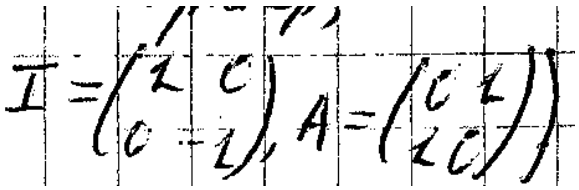
(32) Example:

$$M_1 = I\frac{\partial}{\partial \lambda} + \frac{A}{\lambda}, \qquad I^2 = 1, \quad A^2 = -1,$$
$$IA + AI = 0;$$
$$M_2 = M_1\frac{\partial}{\partial \lambda} + \frac{C}{\lambda^2}, \qquad C = -\frac{1}{2}(I + A).$$

[To the right, beyond a vertical rule:]

(I, A — matrices, for example,

$$I = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix})$$



Original source details: GELFAND-P001362

Uncertain reading. The off-diagonal signs in A have the same shape as the ones in I.

$$L_1 = \frac{\partial}{\partial x_1} + M_1, \qquad L_2 = \frac{\partial}{\partial x_2} + M_1^2, \qquad L_3 = \frac{\partial}{\partial x_3} + M_2.$$

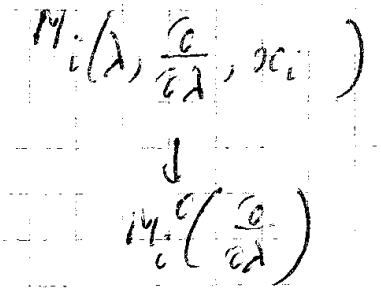
A great deal of computation is needed.

Finding solutions of the equations:

$$L_i y = 0. \tag{33}$$

GELFAND-P001363

Begin with the G.–L.–M. integral equations in λ, take n-soliton solutions, and degenerate them to rational solutions.



Original source details: GELFAND-P001363

Diagram and typed description.

$$M_i(\lambda, \tfrac{\partial}{\partial \lambda}, x_i)$$
$$\downarrow$$
$$M_i^0(\tfrac{\partial}{\partial \lambda})$$

construct $K(\lambda, \lambda', x_i)$, $F(\lambda, \lambda', x_i)$

and then it will remain (possibly “it remained”)[?] to construct rational solutions.



Uncertain reading. it will remain (possibly “it remained”)

Original source details: GELFAND-P001363

This includes dissipative systems too.

The next generalization:

$$L_i = R_i\left(\lambda, \frac{\partial}{\partial\lambda}\right)\frac{\partial}{\partial x_i} - M_i\left(\lambda, \frac{\partial}{\partial\lambda}\right). \tag{34}$$

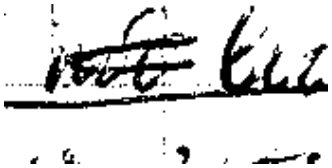


Uncertain reading. The number reads (34) beneath overlapping strokes.

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A. A. Migdal.

Another meaning of [crossed out: illegible start, possibly “вы...”[?]]



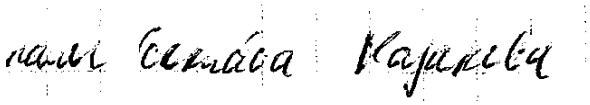
Uncertain reading. illegible start, possibly “вы...”

Original source details: GELFAND-P001363

calculating matrix integrals and the passage
from matrix integrals to continuum integrals

[“Matrix” is inserted above the line before “integrals”.]

A noncommutative analogue of Karasev’s symbol (likely reading)[?].



Uncertain reading. of Karasev’s symbol (likely reading)

Original source details: GELFAND-P001363

(Operators from the outset in place of $N \times N$ matrices)

$$\int dA\, \exp\left(-\frac{N}{\lambda}\operatorname{tr} V(A)\right), \qquad A^+ = -A, \quad N \to \infty. \tag{1}$$

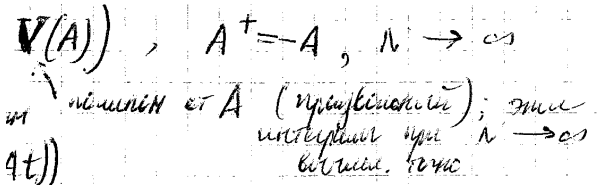


Diagram and typed description.

$V(A)$
\
a polynomial in A (arbitrary); these integrals as $N \to \infty$ are computed exactly

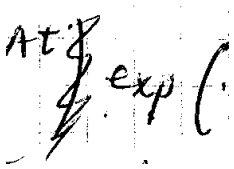
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A way to compute them. consider

$$\psi(t) = \frac{1}{N}\,\mathrm{tr}\big(\exp(At)\big) \tag{2}$$

and derive the equation of motion (Schwinger–Dyson). It is clear that

$$0 = \int dA\,\frac{1}{N^2}\,\mathrm{tr}\left(\frac{\partial}{\partial A}\left[e^{At}\exp\left(-\frac{N}{\lambda}\,\mathrm{tr}\,V(A)\right)\right]\right). \tag{3}$$



Uncertain reading. A bracket or several brackets are crossed out after e^{At}.

Original source details: GELFAND-P001363

(the integral of a total derivative $= 0$). This is equivalent to

$$0 = \int_0^t d\tau\,\psi(\tau)\psi(t-\tau) - \frac{1}{\lambda}V'\left(\frac{\partial}{\partial t}\right)\psi(t), \tag{4}$$

$$R(\omega) = \psi^2(\omega) - \frac{1}{\lambda}V'(\omega)\psi(\omega). \tag{5}$$

What is of interest is the average defined by the integral, $\langle f(A)\rangle$

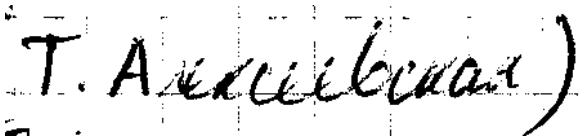
$$\langle f(A)\rangle = \frac{\int dA\,\exp(\ldots)f(A)}{\int dA\,\exp(\ldots)} \tag{6}$$

[On the right:]
($f(A)$ — some invariant function —
— a function of the eigenvalues.

GELFAND-P001364

[source number: 26]

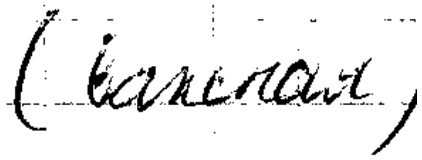
I. M. G. Problem. (T. Alekseevskaya / Aleksievskaya?)



Uncertain reading. Alekseevskaya / Aleksievskaya

Original source details: GELFAND-P001364

(uncertain reading, possibly “solved it”?)



Uncertain reading. uncertain reading, possibly “solved it”

Original source details: GELFAND-P001364

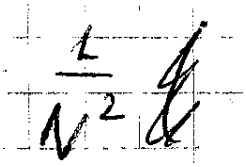
Given a cube (3-dimensional). Draw a plane through every three of its vertices. Into how many pieces do they cut the cube?

(Answer: 96)

The limit as $N \to \infty$ is the same for all the classical groups:

$$O(\infty) = SU(\infty) = SP(\infty). \tag{7}$$

$$0 = \int dA \, \frac{1}{N^2} \frac{\partial}{\partial A_{ij}} \left\{ (e^{At})^{ji} \exp\left(-\frac{N}{\lambda} \operatorname{tr} V(A) \right) \right\}. \tag{3'}$$



Uncertain reading. A symbol after $1/N^2$ is crossed out, possibly tr.

Original source details: GELFAND-P001364

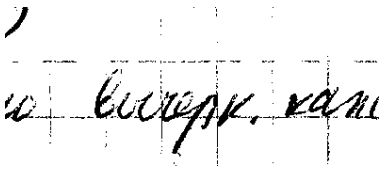
The formula used was:

$$\frac{\partial}{\partial A_{ij}} (e^{At})_{ji} = \int_0^t d\tau \, (e^{A\tau})_{jj} (e^{A(t-\tau)})_{ii}. \tag{8}$$

Derivation: differentiate

$$\prod (1 + A\Delta t) = T \exp\left(\int A \, dt \right)$$

and get the integral (in turn pulling out (likely reading)? each matrix element in its own place)



Uncertain reading. pulling out (likely reading)

Original source details: GELFAND-P001364

Another formula from matrix theory that was used:

$$\frac{\partial}{\partial A_{ij}} \operatorname{Tr} V(A) = (V'(A))_{ji}. \tag{9}$$

From (8) and (9), (4) follows:

$$(e^{At})^{ji} \frac{\partial}{\partial A_{ij}} \operatorname{Tr} V(A) = \\ = (V'(A))_{ji} (e^{At})_{ij} = V'\left(\frac{\partial}{\partial t} \right) \operatorname{tr}(e^{At}).$$

In (4) we should write the average ($\int dA$ is omitted).

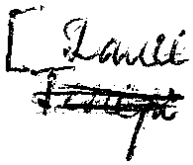
It was said that

$$0 = \int_0^t d\tau \, \langle \psi(\tau) \psi(t-\tau) \rangle - \frac{1}{\lambda} V'\left(\frac{\partial}{\partial t} \right) \langle \psi(t) \rangle. \tag{10}$$

[Next we must consider

$$\int \mathcal{D}A_\mu \exp\left(+\int d^4x \,\frac{N}{\lambda}\,\mathrm{tr}\,F_{\mu\nu}^2\right) \tag{11}$$

]



Uncertain reading. "Next" is inserted above a crossed-out word, probably "Now".

Original source details: GELFAND-P001365

We use the formula

$$\left\langle \frac{\mathrm{tr}\,A}{N}\frac{\mathrm{tr}\,B}{N}\right\rangle = \left\langle \frac{\mathrm{tr}\,A}{N}\right\rangle\left\langle \frac{\mathrm{tr}\,B}{N}\right\rangle + O\left(\frac{1}{N^2}\right). \tag{12}$$

Applying it, we obtain

$$0 = \int_0^t d\tau\,\langle\psi(\tau)\rangle\langle\psi(t-\tau)\rangle - \frac{1}{\lambda}V'\left(\frac{\partial}{\partial t}\right)\langle\psi(t)\rangle. \tag{13}$$

Now introduce the Laplace transform:

$$f(\omega) = \int_0^\infty dt\, e^{-\omega t}\langle\psi(t)\rangle; \tag{14}$$

then from (13) (convolution!)
[crossed out: and] we obtain the equation

$$\underbrace{R(\omega)}_{\text{polynomial in }\omega} = f^2(\omega) - \frac{1}{\lambda}V'(\omega)f(\omega) \tag{15}$$

or (taking the jumps):

$$0 = 2\,\mathrm{Im}\,f\left(\mathrm{Re}\,f - \frac{1}{2\lambda}V'(\omega)\right). \tag{16}$$

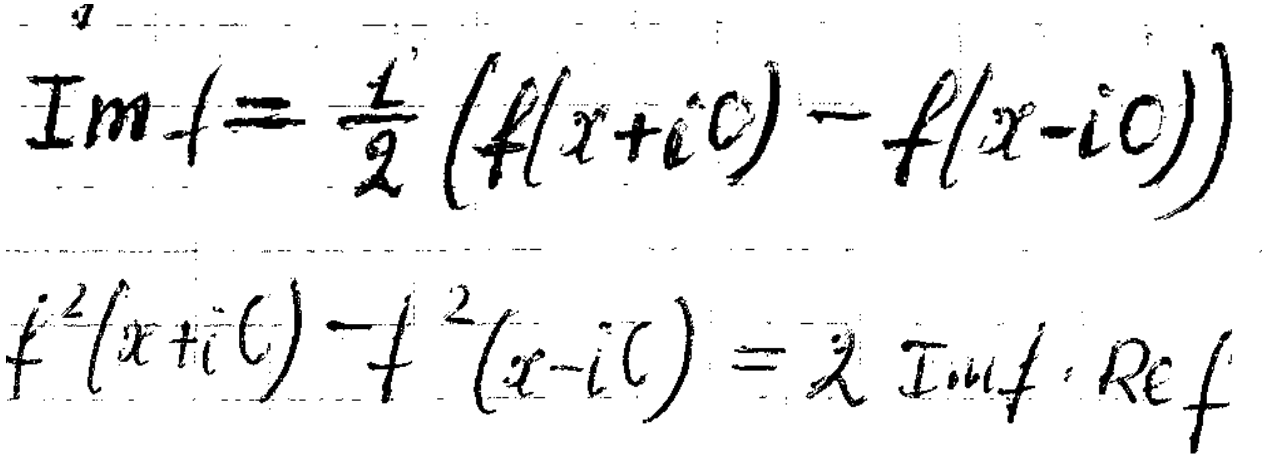
Here

$$\operatorname{Im} f = \frac{1}{2}\big(f(x+i0) - f(x-i0)\big). \tag{17}$$

But it is easy to see that

$$f(\omega) = \frac{1}{N}\left\langle \operatorname{tr} \frac{1}{\omega - A} \right\rangle. \tag{18}$$

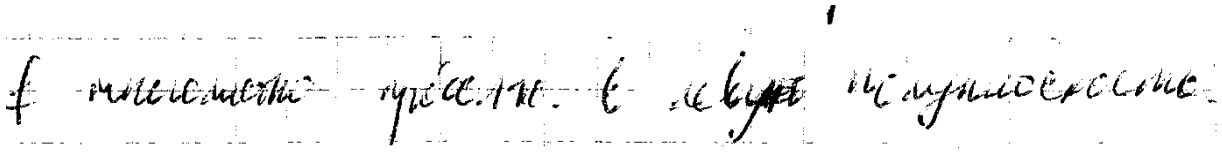
$$f^2(x+i0) - f^2(x-i0) = 2 \operatorname{Im} f \cdot \operatorname{Re} f. \tag{19}$$



Original source details: GELFAND-P001365

Uncertain reading. Equation (17) shows $1/2$, and (19) has a factor of 2. There is a faint separate stroke above the numerator in (17).

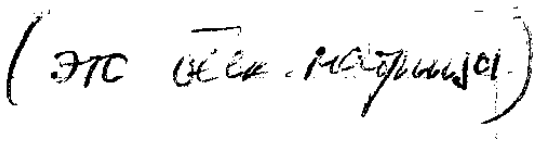
(20) f is fully represented in the left half-plane.



Original source details: GELFAND-P001365

Uncertain reading. The reading “is fully represented” is uncertain. The phrase may refer to analytic continuation.

$f(\omega)$ has jumps at the points in the spectrum of the operator A (this is an infinite matrix (likely reading)[?])



Uncertain reading. this is an infinite matrix (likely reading)

Original source details: GELFAND-P001365

GELFAND-P001366

[source number: 27]

Now we obtain the old (French) equation

$$f(\omega) = \int \frac{\varrho(\lambda)\, d\lambda}{\omega - \lambda}. \tag{21}$$

$$f = \frac{1}{2\lambda} V'(\omega) + M(\omega)\sqrt{(\omega - a)(\omega - b)}. \tag{22}$$

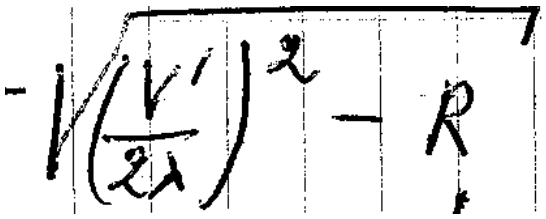
[Zakharevich.

$$A(x)(B(x) - \alpha(x)) = 0, \qquad iA(x) + B(x) \quad \text{continues analytically to } \operatorname{Im} x > 0$$

and so on.]

Instead one can solve the quadratic equation (15):

$$f = \frac{1}{2\lambda} V'(\omega) - \sqrt{\left(\frac{V'}{2\lambda}\right)^2 - R}. \tag{23}$$



Uncertain reading. In (23), the term under the square root is written $-R$.

Original source details: GELFAND-P001366

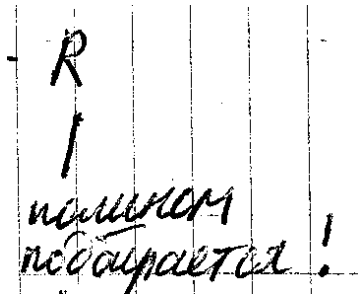
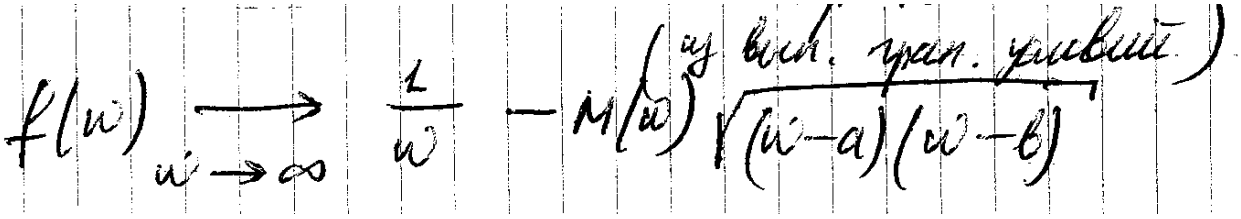


Diagram and typed description.

R
|
the polynomial
is chosen!

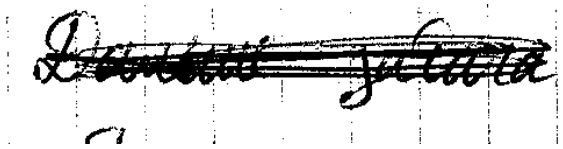
Original source details: GELFAND-P001366

$$f(\omega) \underset{\omega \to \infty}{\longrightarrow} \frac{1}{\omega} \quad \overset{\text{(from satisfying the boundary conditions)}}{-M(\omega)\sqrt{(\omega - a)(\omega - b)}} \tag{24}$$



Original source details: GELFAND-P001366

Uncertain reading. After $1/\omega$ in (24), the manuscript has $-M(\omega)\sqrt{(\omega - a)(\omega - b)}$.



Original source details: GELFAND-P001366

Uncertain reading. A crossed-out line, possibly “Next, the problem” or “The given equation”.

For field theory, Karasev’s method does not go through:

[crossed out: unclear start, possibly “ур” or “при”?] the components do not commute and cannot be put simultaneously in diagonal form.



Original source details: GELFAND-P001366

Uncertain reading. unclear start, possibly “ур” or “при”

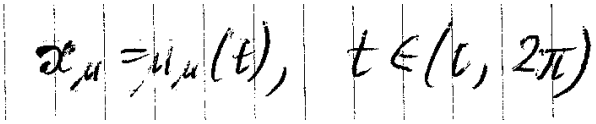
Consider the integral (11).

$$A_\mu(x) : \longrightarrow\ W(c) = \frac{1}{N} \operatorname{tr} P \exp\left(\oint_c A_\mu\, dx_\mu\right) \tag{25}$$

and repeat the same reasoning

$F_{\mu\nu}$ is a tensor in μ, ν and a matrix in the Lie algebra of $SU(N)$.

$$x_\mu = u_\mu(t), \qquad t \in (0, 2\pi)$$



Original source details: GELFAND-P001366

Uncertain reading. The right side more likely reads $u_\mu(t)$. Compare x_μ on the left.

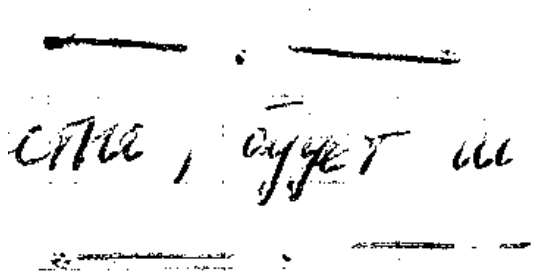
$$\int A_\mu(x(t))\dot{x}_\mu\, dt$$

$\sim \exp At$ earlier

A chain of identities: . . .

GELFAND-P001367

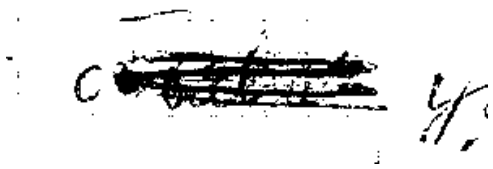
$$\int \mathcal{D}A\, \hat{P}\, \frac{\partial}{\partial A} \exp\left(\oint_c A\, dy\right) \tag{26}$$



Original source details: GELFAND-P001367

Diagram and typed description. Two strokes with a dot between them. Similar faint traces appear below the announcement.

30 March it is uncertain whether the seminar will take place (if cancelled, then [crossed out: initial “c” visible, remainder illegible[?]] ask at the lab 139-53-57)



Uncertain reading. initial “c” visible, remainder illegible

Original source details: GELFAND-P001367

GELFAND-P001368

13/IV-87 Gelfand seminar

Sinai

On spin glasses (following Parisi)

$$H = \sum_{\langle i,j \rangle} J_{ij}\sigma_i\sigma_j; \qquad J_{ij} \text{ are random.} \tag{1}$$

(Ising model of a spin glass)

$\langle i, j\rangle$ — a pair of neighbors

Spin glass: when the interaction forces are long-range and oscillate strongly

[crossed out: short entry, possibly “Par...” or “Theor...”[?]]



Uncertain reading. short entry, possibly “Par...” or “Theor...”

Original source details: GELFAND-P001368

Parisi: everything interacts, namely:

$$H = \frac{1}{\sqrt{N}} \sum J_{ij}\sigma_i\sigma_j \tag{2}$$

[On the right:] J_{ij} — a Gaussian random variable (all with the same variance)

There are many ground states (differing in energy by $\sim 1/N$);

$$\sigma_i = \pm 1.$$

One must study the case

$$N \to \infty \qquad \text{and} \qquad \lim_{N\to\infty} \min H$$
$$\text{is computed}$$

Entropy of the set of [inserted above the line: (ground)] configurations > 0 (in the limit $N \to \infty$)

$$\frac{\ln \#[O(N)]}{N} \underset{N\to\infty}{\longrightarrow} h > 0. \tag{3}$$

The property of ultrametricity for ground states (Parisi, Virasoro): if $\sigma', \sigma'', \sigma'''$ — are three ground states, then among the numbers

$$\frac{1}{N}(\sigma', \sigma''), \quad \frac{1}{N}(\sigma', \sigma'''), \quad \frac{1}{N}(\sigma'', \sigma''') \tag{4}$$

two are necessarily equal, and the third is greater.

GELFAND-P001369

Phase transition to a spin glass.

Parisi invented stochastic quantization.

A. S. Schwarz.

Superconformal transformations and a superconformal manifold.

There is a very complete and beautiful generalization of complex analysis to $(1,1)$-dimensional superconformal manifolds.

Definition

Let there be a superdomain (x,θ). Introduce the covariant derivative:

$$D = \frac{\partial}{\partial\theta} + \theta\frac{\partial}{\partial x}. \tag{1}$$

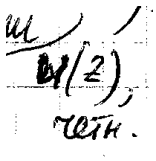
In dimension $(1,1)$.

A map $(x,\theta) \to (\widetilde{x},\widetilde{\theta})$ is called superconformal, if

$$\widetilde{D} = \underset{\text{function}}{F} D$$

(D is preserved up to a factor); here $\widetilde{D} = \frac{\partial}{\partial\widetilde{\theta}} + \widetilde{\theta}\frac{\partial}{\partial\widetilde{x}}$.

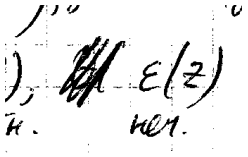
They are parametrized [inserted: by pairs]



Uncertain reading. $w(z)$, possibly $u(z)$

Original source details: GELFAND-P001369

[caption: even], [crossed out: illegible sign?]



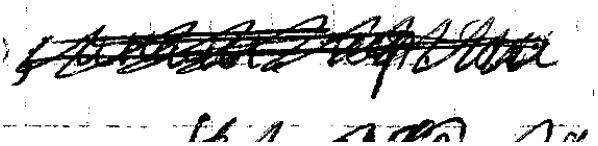
Uncertain reading. illegible sign

Original source details: GELFAND-P001369

$\varepsilon(z)$ [caption: odd] — analytic functions.

Superconformal manifold: glued from such superdomains.

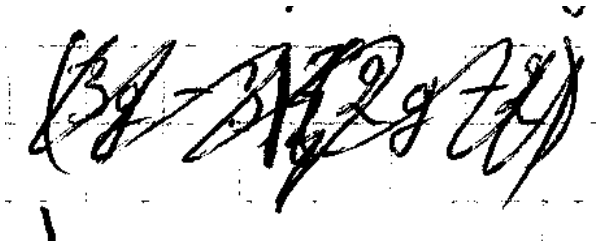
For them one can construct a theory [crossed out: illegible text?]



Uncertain reading. illegible text

Original source details: GELFAND-P001369

of superconformal curves, the moduli space has dimension [crossed out: $(3g-3 \mid 2g+2)$?[?]]



Uncertain reading. $(3g-3 \mid 2g+2)$?

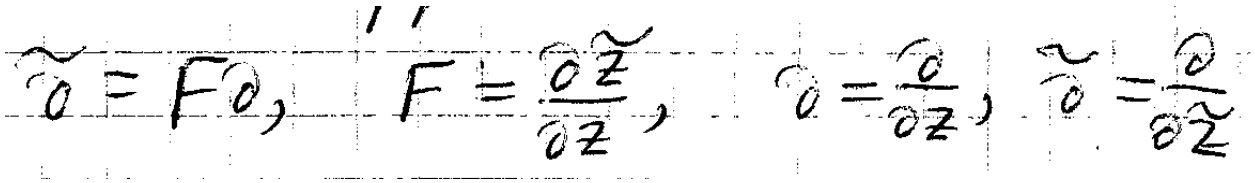
Original source details: GELFAND-P001369

$(3g-3 \mid 2g-2)$ (g — genus of the underlying surface).

There is an analogue of Abelian differentials and higher differentials.

For ordinary conformal maps:

$$\widetilde{\partial} = F\partial, \qquad F = \frac{\partial \widetilde{z}}{\partial z}, \qquad \partial = \frac{\partial}{\partial z}, \qquad \widetilde{\partial} = \frac{\partial}{\partial \widetilde{z}}.$$



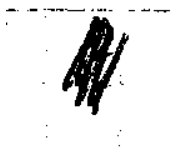
Original source details: GELFAND-P001369

Uncertain reading. The manuscript gives $\widetilde{\partial} = F\partial$ together with $F = \partial\widetilde{z}/\partial z$.

Question by I. M. G.: an analogue of the Riemann mapping theorem?

Answer (A. S. S.): if the underlying space is simply connected, it reduces to the standard one.

Question by I. M. G.: description of the ring of functions [crossed out: illegible sign[?]]

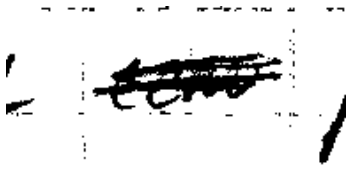


Uncertain reading. illegible sign

Original source details: GELFAND-P001369

on a compact superconformal manifold.

Answer (A. S. S.): there is a Riemann–Roch theorem if [crossed out: illegible word[?]]



Uncertain reading. illegible word

Original source details: GELFAND-P001369

the dimensions of all the objects make sense.

(2) Def. Superspace: a free Λ-module, where Λ is a Grassmann algebra.

$(x_1, \ldots, x_n, \theta_1, \ldots, \theta_n)$ — even and odd elements $\Lambda = \Lambda(e_1, \ldots, e_k)$.

GELFAND-P001370

[source number: 49]

Specify a linear equation in this module. It singles out a submodule.

$$\sum a_k V_k = 0$$



Uncertain reading. Below “equation”: $\sum a_k V_k = 0$. The letter reads V or v.

Original source details: GELFAND-P001370

It will not be free and will not have [crossed out: illegible sign[?]] a dimension.

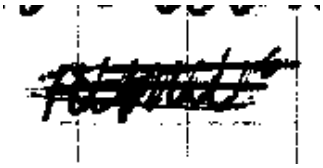


Uncertain reading. illegible sign

Original source details: GELFAND-P001370

[between the lines: functions with specified singularities may have no dimension.]

But in a [crossed out: possibly “general”[?]] typical situation it is a free module, and there is a standard form of the Riemann–Roch theorem.



Uncertain reading. possibly “general”

Original source details: GELFAND-P001370

Abelian differentials.



Uncertain reading. In the narrow left margin, a mark resembling e, 2, or ε.

Original source details: GELFAND-P001370

If Φ is a scalar function, then $D\Phi$ transforms accordingly in different local coordinates.

Fields that transform in this way, sections of Ω, are analogues of Abelian differentials. [above the line: sections of the bundle Ω.]

Suppose there is an Abelian differential V.

For a simply connected underlying surface, $V = D\Phi$.

Period: take the universal cover, where there is $\overline{\Phi}$, shifts, and so on.

There is a Jacobian, a Picard manifold, and so on.

The string partition function is expressed through an analogue of the Selberg ζ-function (joint work with Manin).

———— · ————

Fermionic string $g_{\alpha,\beta};\ X^a$.

Superanalogue of the Riemannian metric: an odd vector field $\widehat{e}$, for which

$$\{\widehat{e}, \overline{\widehat{e}}\} = \tau\widehat{e} + \overline{\tau}\,\overline{\widehat{e}}. \tag{3}$$

(4) Example: $\widehat{e} = \lambda D$.

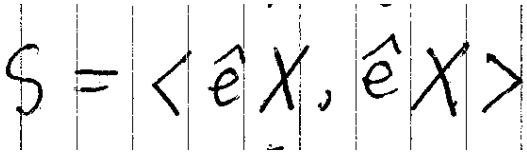
(5) Assertion. There always $\exists$ exists a coordinate system in which the super-Riemannian metric has the form (4).

Explanation: an ordinary Riemannian metric is given by the field $\lambda\frac{\partial}{\partial z}$, λ is determined up to the phase $e^{i\alpha}$.

(This is for any $(2 \mid 2)$-dimensional manifold.)

Fermionic string

$$S = \langle \widehat{e}X, \widehat{e}X \rangle. \tag{6}$$



Original source details: GELFAND-P001370

Uncertain reading. In (6) both e's carry a hat. The bar used in (3) is absent.

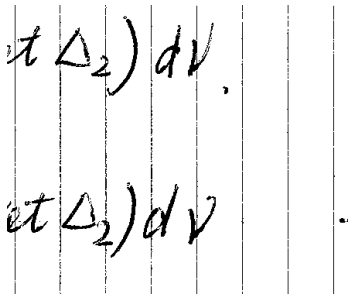


Original source details: GELFAND-P001370

Uncertain reading. An illegible sign below (6) is crossed out.

$$\int (\det \Delta_0)^{-13} (\det \Delta_2)\, dV, \qquad \text{for the bosonic string.} \tag{7}$$

$$\int (\det \Delta_0)^{-5} (\det \Delta_2)\, dV \qquad \text{— for the fermionic string.} \tag{8}$$



Uncertain reading. The differential in (7) and (8) reads dV or $d\nu$.

Original source details: GELFAND-P001370

The measure (7) has asymptotic properties (Belavin–Knizhnik).

GELFAND-P001371

Baranov, Schwarz: extension to the fermionic case, using the work of F.[?] Voronov



Uncertain reading. The initial resembles Cyrillic «Ф.»

Original source details: GELFAND-P001371

(reads as “supermanif.”[?]; computation of measures).



Uncertain reading. reads as “supermanif.”

Original source details: GELFAND-P001371

Another topic:

A noncommutative analogue of Ray–Singer torsion.

(no promise yet.)

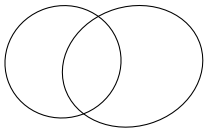
Alekseevskii, D.

Quaternionic manifolds

Let



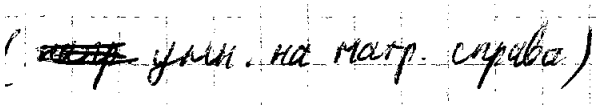
Diagram and typed description.



Original source details: GELFAND-P001371

$$U \subset \mathbb{H}^n, \qquad V \subset \mathbb{H}^n$$
$$f : U \to V$$
$$\forall x \in U \qquad df_x : \mathbb{H}^n \to \mathbb{H}^n \tag{1}$$

a morphism of left modules ([crossed out: possibly "matr."?]



Uncertain reading. possibly "matr."

Original source details: GELFAND-P001371

multiplication by a matrix on the right).

If it is simply connected, only a linear space results. [Below the line: "(".]

All automorphisms are linear.

One can write down the Lie algebra of infinitesimal transformations preserving it.

(2) Assertion: then the transition functions must be linear

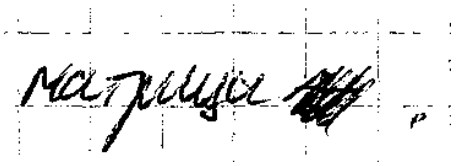
The first approximation "is trivial," with the Russian ending partly illegible[?].



Uncertain reading. "is trivial," with the Russian ending partly illegible

Original source details: GELFAND-P001371

A vector field whose differential is a quaternionic matrix [crossed out: illegible mark[?]].



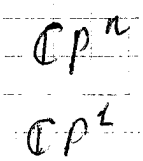
Uncertain reading. illegible mark

Original source details: GELFAND-P001371

Definition (1) is incorrect, since the projective space would then have no quaternionic structure.

$\mathbb{C}P^n$ — the space of one-dimensional quaternionic subspaces of $\mathbb{H}^{n+1}$.

$\mathbb{C}P^1$ — pairs (λ_1, λ_2) up to left multiplication $(\lambda\lambda_1, \lambda\lambda_2)$.



Uncertain reading. Both projective-space symbols visibly use the letter $\mathbb{C}$.

Original source details: GELFAND-P001371

2 charts

$$(\lambda_1, 1)$$
$$(1, \lambda_2)$$
$$(\lambda_1, 1) \sim (1, \lambda_1^{-1})$$

$\lambda \to \lambda^{-1}$ — the transition function

GELFAND-P001372

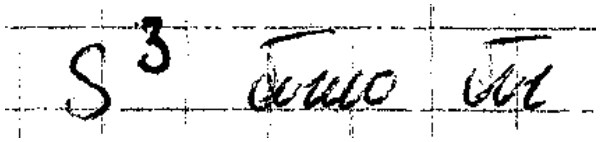
[source number: 50]

λ^{-1}

$$(\lambda_0 + i\lambda_1 + j\lambda_2 + k\lambda_3)^{-1} = \frac{\lambda_0 - i\lambda_1 - j\lambda_2 - k\lambda_3}{\lambda_0^2 + \lambda_1^2 + \lambda_2^2 + \lambda_3^2}.$$

$$d(\lambda^{-1}) = -\lambda^{-1} \cdot d\lambda \cdot \lambda^{-1}$$

Topological explanation: if it existed, then S^3 would have 3 complex structures.



Uncertain reading. The superscript on S reads 3.

Original source details: GELFAND-P001372

$$V = \mathbb{R}^{4n}$$

(3) Quaternionic structure — a subalgebra $Q \subset$ [crossed out: illegible mark?]



Uncertain reading. illegible mark

Original source details: GELFAND-P001372

End(V), isomorphic to $\mathbb{H}$.

(Or a subalgebra spanned by 3 anticommuting complex structures

$$Q = \langle I, J, K, E \rangle \qquad I^2 = J^2 = K^2 = -1, \quad IJ = -JI = K$$

and so on.

$\Rightarrow$ one can multiply on the left by quaternions

Nuance: The isomorphism is unspecified. There is no need to single out I, J and K.

($\mathbb{H}$ has nontrivial automorphisms)

Quaternionic structure on a manifold: a field of quaternionic structures (in each tangent space) (Or: an almost quaternionic structure)

Chern (1945) considered such structures

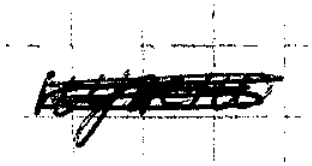
($\forall$ 4-dimensional Riemannian manifold has two, since

$$\mathrm{SO}(4) = \text{product}.$$

$$M^{4n}, \quad Q\colon x \mapsto Q_x$$

Integrability: Cauchy–Riemann equations in place of an atlas

(Up to second-order small terms [crossed out: illegible word?]



Uncertain reading. illegible word

Original source details: GELFAND-P001372

like a quaternionic transformation)

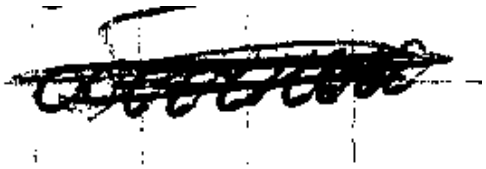
Automorphisms preserving the quaternionic structure

$$G = \mathrm{GL}(1, \mathbb{H}) \cdot \mathrm{GL}(n, \mathbb{H})$$

— group of automorphisms of the quaternionic structure

$$G = \mathrm{Sp}(1) \cdot \mathrm{Sp}(n)$$

— for [crossed out: possibly “ordinary”?]



Original source details: GELFAND-P001372

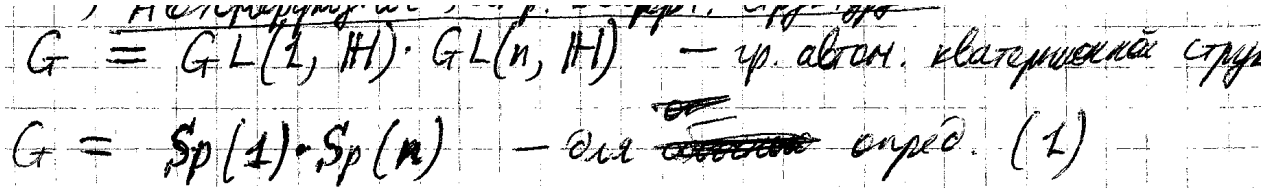
Uncertain reading. possibly “ordinary”

definition (1). [Also crossed out above: illegible mark?.]



Original source details: GELFAND-P001372

Uncertain reading. illegible mark



Original source details: GELFAND-P001372

Uncertain reading. Both formulas begin with $G =$ in the manuscript. The second line refers to (1).

$$y_i = \sum x_k a_{ki}$$

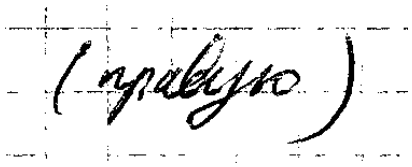
[Crossed out between $\sum$ and x_k: illegible symbol?.]



Original source details: GELFAND-P001372

Uncertain reading. illegible symbol

$\|$ preserves the quaternionic structure (right).



Original source details: GELFAND-P001372

Uncertain reading. The word in parentheses reads “right.”

$$(x_1, \ldots, x_n) \to (\lambda x_1, \ldots, \lambda x_n)$$

also preserves

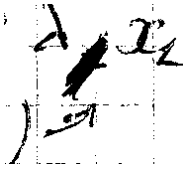
[Crossed out before the first λ in the right-hand tuple: x_1?$^?$.



Original source details: GELFAND-P001372

Uncertain reading. x_1?

Crossed out between this λ and x_1: illegible mark$^?$.]



Original source details: GELFAND-P001372

Uncertain reading. illegible mark

[Below the formula, with an arrow pointing between the first λ and x_1:] (On the other side!)

GELFAND-P001373

Condition for 1-integrability:

$\exists$ a connection ∇ without torsion, preserving the quaternionic structure.

(Linear connection, not necessarily metric.)

[Crossed out in parentheses: An arrow $\to$ is visible. The words on either side may begin "integr..." and "preserv...".$^?$]



Original source details: GELFAND-P001373

Uncertain reading. An arrow $\to$ is visible. The words on either side may begin "integr..." and "preserv...".

Or: $\nabla Q = 0$.

Quaternionic subspace $\longrightarrow$ quaternionic subspace under parallel transport.

Goncharov, A.

Suppose there is a real-analytic manifold with an [inserted above: almost] quaternionic structure.

$$M^{4n}_{\mathbb{R}} \subset M^{4n}_{\mathbb{C}}$$

$$T_x M^{4n}_{\mathbb{C}} \supset K_x = \bigcup \{v + \sqrt{-1}\, Jv\}, \qquad v \in T M^{4n}_{\mathbb{R}}$$

[Above K_x: cone]

$$J \in Q_x, \qquad J^2 = -1$$

And there must be integrability.

D. A.

$$\gamma \text{ — a curve} \quad \Rightarrow \quad \tau_\gamma Q_{\gamma(0)} = Q_{\gamma(1)}$$

$$\gamma : [0,1] \to M$$

(τ_γ — parallel transport along γ)

Q_x — a subalgebra in Aut $T_x M$, it too can be transported.



Uncertain reading. The manuscript reads "a subalgebra in Aut $T_x M$."

Original source details: GELFAND-P001373

GELFAND-P001374

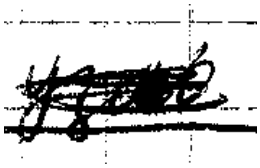
20/IV-87 Gelfand Seminar.

I.M.G. Question: there are 2 commuting isometries in an n-dimensional space. [Added above the line, with a pointer: not defined everywhere.] How can one turn them into 2 commuting unitary operators? What are the obstructions?

Kasparov has partly done this.

Kirillov on Jones's work.

Distinguishing [crossed out: possibly "knots"?]



Uncertain reading. possibly "knots"

Original source details: GELFAND-P001374

knots using factors

Consider the algebra generated by the generators T_i (infinitely many of them) with the relations

$$\begin{cases} (T_i+1)(T_i-q)=0, \\ T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \\ T_iT_j = T_jT_i, \quad |i-j|>1 \end{cases} \tag{1}$$

(Hecke algebra of series A)

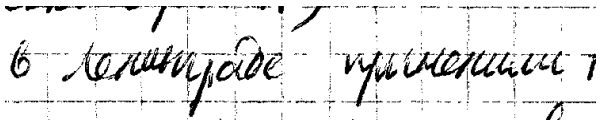
$\Rightarrow$ [crossed out: illegible mark?]

Uncertain reading. illegible mark



Original source details: GELFAND-P001374

in Leningrad they likewise applied series B.



Uncertain reading. The verb reads «применили» (“applied”).

Original source details: GELFAND-P001374

1. Knot $\longrightarrow$ polynomial, equivalent knots $\longrightarrow$ equal polynomials.

2. Factor $\longrightarrow$ subfactor

GELFAND-P001375

Consider $\bigsqcup_n B_n$ ([crossed out: illegible abbreviation?]

Uncertain reading. illegible abbreviation

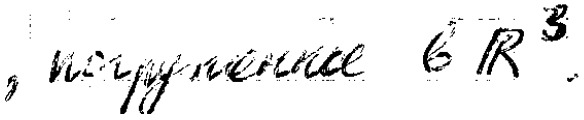


Original source details: GELFAND-P001375

disjoint union). [Below B_n: braid group.]

Link: an oriented compact 1-dimensional manifold immersed in $\mathbb{R}^3$.

Uncertain reading. The definition uses “immersed.”

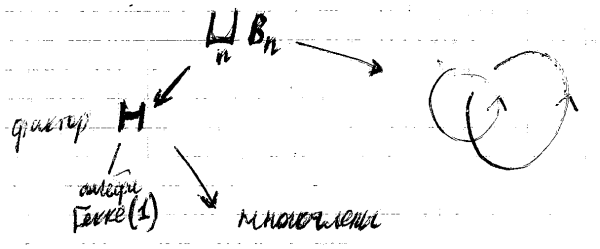


Original source details: GELFAND-P001375

Uncertain reading. A short isolated stroke to the left of the diagram.

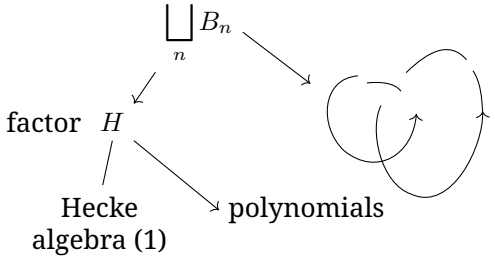


Original source details: GELFAND-P001375



Original source details: GELFAND-P001375

Diagram and typed description.

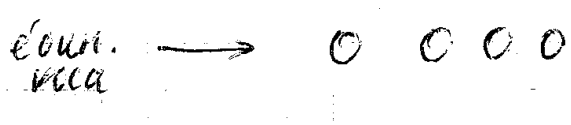




Original source details: GELFAND-P001375

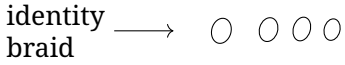
Uncertain reading. Both visible arrowheads point upward along the right branches. The upper arcs are broken or faint. Their joins and the upper crossing remain uncertain.

Braid $\longrightarrow$ link (join the beginning to the end).

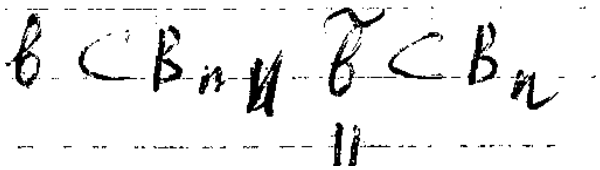


Original source details: GELFAND-P001375

Diagram and typed description.



(2) $b \subset B_n$ and $\widetilde{b} \subset B_n$ give equivalent links. [Below $\widetilde{b}$:] $\widetilde{b} = cbc^{-1}$. [To the right of the vertical line:] $B_n \subset B_{n+1}$.



Original source details: GELFAND-P001375

Uncertain reading. In (2) the signs preceding B_n look like $\subset$.

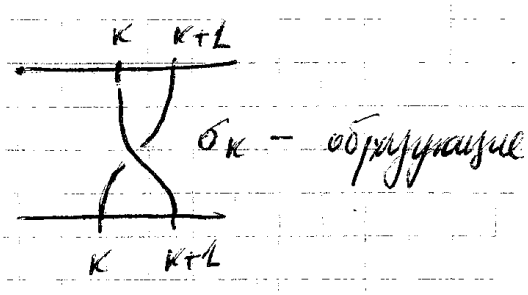
(3) $b \in B_n$, $\widetilde{b} \in B_{n+1}$, $\widetilde{b} = b \cdot \sigma_n^{\pm 1}$ (appending one generator from B_{n+1}).
Then b and $\widetilde{b}$ produce [crossed out: possibly "equiv."?]



Original source details: GELFAND-P001375

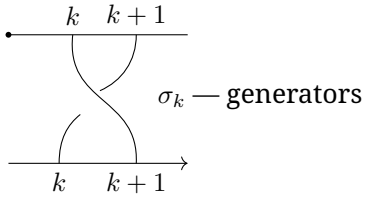
Uncertain reading. possibly "equiv."

equivalent links.



Original source details: GELFAND-P001375

Diagram and typed description.



(4) Markov's theorem (difficult). The operations described suffice to pass to any equivalent link.

Idea: construct a function invariant under these operations.

The map $\bigsqcup_n B_n$ into H is constructed as follows: $\sigma_i \to T_i$, $\sigma_i^{-1} \to T_i^{-1}$. [Below T_i^{-1}: found from the quadratic equation $(1)_1$.]

$(1)_{2,3}$ — a complete set of relations in the braid group.

$$\sigma_i^{-1} \longrightarrow \frac{T_i + (1-q)}{q} = T_i^{-1}$$

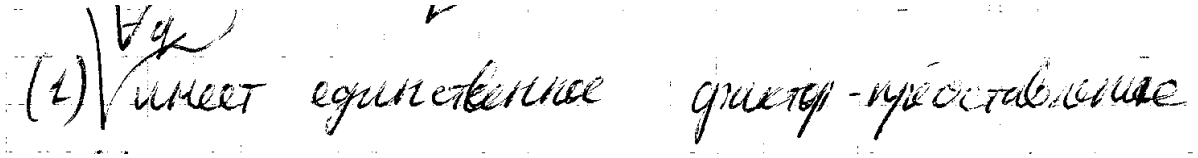
The set of generators (1) [inserted above: $\forall q$] has a unique factor representation with a trace, and [crossed out: illegible fragment?]

Uncertain reading. illegible fragment



Original source details: GELFAND-P001375

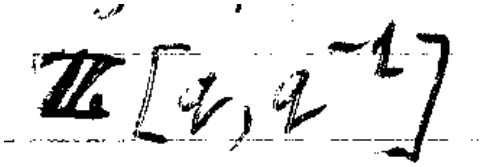
the trace is a polynomial in q, q^{-1}.



Original source details: GELFAND-P001375

Uncertain reading. The manuscript inserts $\forall q$ above the assertion.

$$\mathbb{Z}[z, z^{-1}]$$



Uncertain reading. The variable in the bottom formula resembles z or v. The q in the preceding line has a different shape.

Original source details: GELFAND-P001375

GELFAND-P001376

[source number: 57]

Exercise. Take $q = \pm 1$ and see what invariant of the braid group results (the ratio of the values of the Jones polynomial at these values).

Jones specifically excludes from consideration [crossed out: illegible fragment?]

Uncertain reading. illegible fragment



Original source details: GELFAND-P001376

roots of 1.

I.M.G.'s idea: consider precisely the roots of 1. It may turn out that there are no factors at all and that roots of 1 are what is needed.

A.M. Vershik has already done something on this.

A. Beilinson.

On work by Beilinson, MacPherson and V. Schechtman.

Notes on motivic cohomology

Let X — an analytic manifold, E — a vector bundle on X. [Under the description of X:] (or "complex," abbreviated[?])



Uncertain reading. "complex," abbreviated

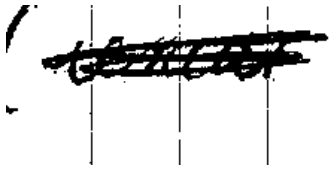
Original source details: GELFAND-P001376

(compact).

$$c_i(E) \in H^{2i}(X, \mathbb{C}) = H^{2i}(X, \mathcal{O} \to \Omega^1 \to \ldots)$$

[Above the final argument, with a pointer: cohomology with coefficients in a complex of sheaves.]

They are integral and satisfy the Hodge condition ([crossed out: illegible word[?]]



Uncertain reading. illegible word

Original source details: GELFAND-P001376

$c_i \in i$th term of the Hodge filtration).

(Since they can be computed via log derivatives in Čech cocycles.)



Uncertain reading. The word following log is abbreviated «произв.», read here as "derivatives."

Original source details: GELFAND-P001376

We will compute the manifold's cohomology from the de Rham complex.

$$0 \to \mathbb{C} \to \mathcal{O} \to \Omega^1 \to \Omega^2 \to \ldots \tag{1}$$

[In (1), between $\mathbb{C} \to$ and $\mathcal{O}$ crossed out: illegible mark[?].]



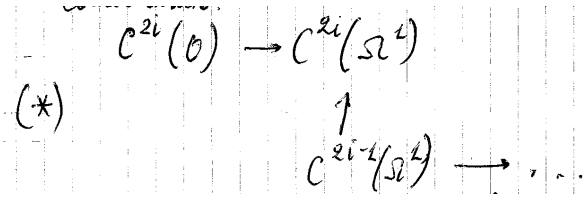
Uncertain reading. illegible mark

Original source details: GELFAND-P001376

[At right:] Ω^i — holomorphic i-forms (a sheaf).

$\{U_i\}$ — a cover of X by small open sets.

Compatibility:



Original source details: GELFAND-P001376

Diagram and typed description.

$$\begin{array}{ccc} & C^{2i}(\mathcal{O}) & \longrightarrow C^{2i}(\Omega^1) \\ (*) & & \uparrow \\ & C^{2i-1}(\Omega^1) & \longrightarrow \cdots \end{array}$$

Hodge filtration on $H^k(X, \mathbb{C})$

$$F^j H^k(X, \mathbb{C}) = H^k(X, 0 \to \ldots \to 0 \to \Omega^j \to \Omega^{j+1} \to \ldots)$$

Atiyah's construction of the characteristic class, from which it follows that $c_j \in F^j$

g_{ij} — transition functions

$$g_{ij} \colon U_i \cap U_j \longrightarrow \mathrm{GL}_n(\mathbb{C})$$

$$A = g_{ij}^{-1} dg_{ij} \in \Omega^1(U_i \cap U_j, \mathrm{Mat}_n(\mathbb{C}))$$

$$\in C^1(X, \Omega^1 \otimes \mathrm{Mat}_n)$$

[Below $g_{ij}^{-1}dg_{ij}$: a 1-form with values in a Lie algebra.]



Original source details: GELFAND-P001376

Uncertain reading. The formula for A has two successive $\in$ signs.

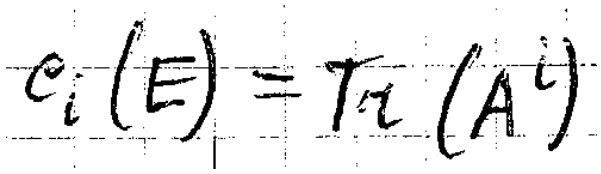
Take $(g_{ij}^{-1}dg_{ij})^k$ — a cochain with values in k-forms. [Below the power: (multiplication of cochains] ‖

Take the trace Tr $\longrightarrow$ the resulting form has one component.

GELFAND-P001377

$$c_i(E) = \mathrm{Tr}(A^i)$$

[At right:] A^i — a form of degree i on the intersection of $(i+1)$ sets.



Original source details: GELFAND-P001377

Uncertain reading. The manuscript writes $c_i(E)$ before the trace of the power.

[Crossed out: possibly «Вид» ("form")?]



Original source details: GELFAND-P001377

Uncertain reading. possibly «Вид» ("form")

$\Rightarrow$ an element of $2i$-dimensional cohomology (see diagram $(*)$).

Problem: explicitly construct a homotopy to an integral characteristic class.

For

$$c_1 = d\log(\det g_{ij})$$

We must drop d:

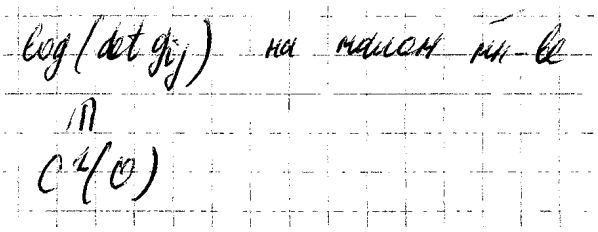


Diagram and typed description.

$\log(\det g_{ij})$ on a small set

$\cap$

$C^1(\mathcal{O})$

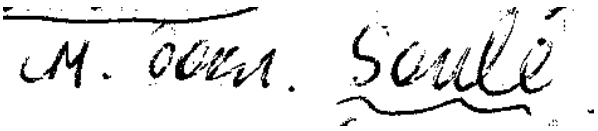
Original source details: GELFAND-P001377

GELFAND-P001378

27.4.87 Gelfand Seminar

A. Beilinson

Deligne cohomology [inserted above the end of the title — Beilinson]: see possibly an abbreviation for "survey"?



Uncertain reading. possibly an abbreviation for "survey"

Original source details: GELFAND-P001378

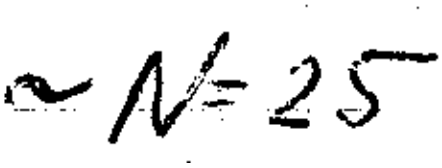
Soulé, Bourbaki Seminar, Feb. 85. [Below the surname:] (Regulateurs).

H. Esnault, E. Viehweg (preprint). [Below "preprint": Manin has it.]

Deligne–Beilinson cohomology

A. Beilinson: Current Problems in Mathematics, vol. 24.

Contempor. math. $\sim$ No. 25



Uncertain reading. After "Contempor. math." there is a $\sim$ sign and what appears to be "No. 25."

Original source details: GELFAND-P001378

Complex of sheaves on any manifold (complex analytic scheme).

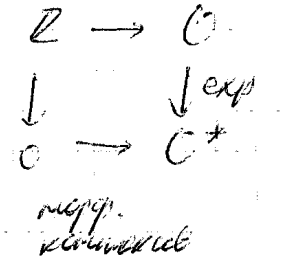
$$\mathbb{Z}(i)_{\mathcal{D}} := \mathbb{Z} \overset{(2\pi\sqrt{-1})^i}{\longrightarrow} \mathcal{O} \longrightarrow \Omega^1 \longrightarrow \ldots \longrightarrow \Omega^{i-1} \tag{1}$$

(i — an integer.)
In particular

$$\mathbb{Z}(0)_{\mathcal{D}} = \mathbb{Z}$$

$$\mathbb{Z}(1)_{\mathcal{D}} = \left(\mathbb{Z} \overset{2\pi\sqrt{-1}}{\longrightarrow} \mathcal{O}\right) \overset{\exp}{\longrightarrow} \underbrace{\mathcal{O}^*}_{\text{degree } 1} [-1]$$

[At right:]



Original source details: GELFAND-P001378

Diagram and typed description.

$$\begin{array}{ccc} \mathbb{Z} & \longrightarrow & \mathcal{O} \\ \downarrow & & \downarrow \exp \\ 0 & \longrightarrow & \mathcal{O}^* \end{array}$$

morphism of complexes

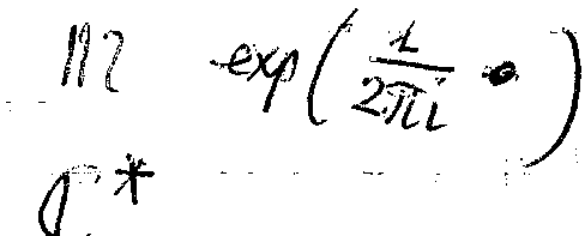
The cohomology sheaf is

$$H^1(\mathbb{Z}(1)) = \mathcal{O}^*$$

$$\mathbb{Z}(2)_{\mathcal{D}} = \left(\mathbb{Z} \overset{-4\pi^2}{\longrightarrow} \mathcal{O} \longrightarrow \Omega^1\right)$$

$$H^1(\mathbb{Z}(2)_{\mathcal{D}}) = \mathbb{C}/(2\pi\sqrt{-1})^2\mathbb{Z}$$

[Below the right-hand side:]



Original source details: GELFAND-P001378

Diagram and typed description.

$$\begin{array}{ccc} \| & & \exp(\frac{1}{2\pi i} \circ) \\ \mathbb{C}^* & & \end{array}$$



Original source details: GELFAND-P001378

Uncertain reading. The last mark in the exponential argument resembles a small circle or dot.

$$H^2(\mathbb{Z}(2)_{\mathcal{D}}) = \Omega^1/d\mathcal{O}$$

GELFAND-P001379

[source number: 67]

There is a map

$$\mathbb{Z}(i)_{\mathcal{D}} \longrightarrow \mathbb{Z} \qquad \text{(component 0)} \tag{2}$$

$$\mathbb{Z}(i)_{\mathcal{D}} \longrightarrow \Omega^i[-i] \tag{3}$$

(0 on all forms, the last one going to its differential)

$$\mathbb{Z}(i)_{\mathcal{D}} \otimes \mathbb{Z}(j)_{\mathcal{D}} \longrightarrow \mathbb{Z}(i+j)_{\mathcal{D}} \tag{4}$$

$$z_i \star \omega_j = z_i \omega_j$$

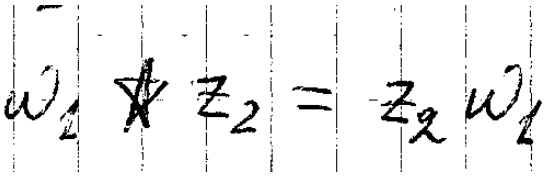
[To the right of (4):] arrows (2) and (3) send this multiplication to ordinary multiplication of numbers and forms.

Let z_j be from component 0.

$$z_1 \star z_2 = z_1 z_2 \qquad \text{(ordinary product.)} \tag{5′}$$

$$z_1 \star \omega_2 = z_1 \omega_2 \tag{5″}$$

$$\omega_1 \star z_2 = z_2 \omega_1 \tag{5‴}$$



Uncertain reading. The right-hand side reads $z_2\omega_1$. A short faint stroke appears before z_2.

Original source details: GELFAND-P001379

$$\omega_1 \star \omega_2 = \begin{cases} 0, & \text{if } \omega_2 \notin \Omega^{j-1}, \\ \omega_1 \wedge d\omega_2, & \text{if } \omega_2 \in \Omega^{j-1} \end{cases} \tag{6}$$

(7) Up to homotopy this multiplication is commutative. [Below «homotopy»: (of chain complexes).]

(8) Both maps (2) and (3) commute with multiplication.

(9) Let E — a holomorphic vector bundle [inserted above: on X]. Then its Chern classes are defined

$$c_i(E) \in H^{2i}(X, \mathbb{Z}(i)_{\mathcal{D}})$$

[At right:] (cohomology group with coefficients in a complex of sheaves). [Added above the beginning of the note: “abelian”?.]



Uncertain reading. “abelian”

Original source details: GELFAND-P001379

(10) Under map (2) they go to the ordinary topological Chern classes. This property and functoriality uniquely determine these classes in (9).

Description of the group $H^{2i}(X, \mathbb{Z}(i)_{\mathcal{D}})$.

Let X be compact.

Čech definition of the cohomology in (9)

C^{ij} — i-dimensional cochains with coefficients in the jth sheaf

[Crossed out to the left of the diagram: illegible mark?.]



Uncertain reading. illegible mark

Original source details: GELFAND-P001379

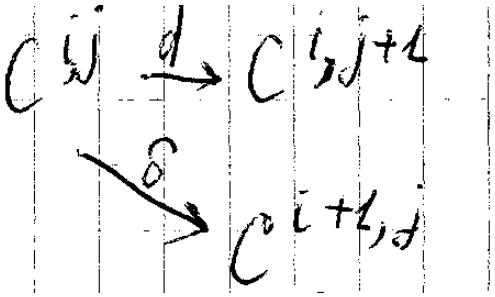


Diagram and typed description.

$$C^{ij} \xrightarrow{d} C^{i,j+1}$$
$$\delta \searrow$$
$$C^{i+1,j}$$

Original source details: GELFAND-P001379

[At right:] d — the differential in the complex (de Rham).

Now for $d + \delta$ take ordinary cohomology.

(These are obtained with C^* [crossed out: illegible mark?]



Uncertain reading. illegible mark

Original source details: GELFAND-P001379

X (integral)).

GELFAND-P001380

For compact X

$$H^n(X,\mathbb{C}) = \bigoplus_{a+b=n} H^{a,b}(X) \qquad \text{(Hodge decomposition)} \tag{11}$$

$$F^i H^n(X,\mathbb{C}) = \bigoplus_{a\geq i} H^{a,b} \qquad \text{(Hodge filtration)} \tag{12}$$

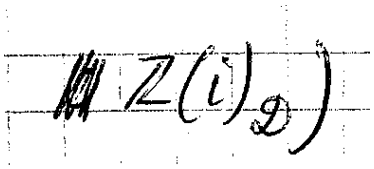
It is known that

$$H^{a,b}(X) = H^b(X,\Omega^a) \tag{13}$$

Then

$$\begin{aligned} 0 \longrightarrow \frac{H^{n-1}(X,\mathbb{C})}{H^{n-1}(X,\mathbb{Z}) + F^i H^{n-1}(X,\mathbb{C})} &\longrightarrow H^n\big(X,\mathbb{Z}(i)_{\mathcal{D}}\big) \\ &\longrightarrow H^n(X,\mathbb{Z}) \cap F^i H^n(X,\mathbb{C}) \longrightarrow 0 \end{aligned} \tag{14}$$

[Crossed out in the middle term of (14), before $\mathbb{Z}(i)_{\mathcal{D}}$: illegible mark?.]



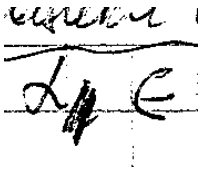
Uncertain reading. illegible mark

Original source details: GELFAND-P001380

Deligne cohomology class $\sim$ (difference of an integral cocycle and a de Rham cocycle) $+$ (a homotopy joining them).

Cochain with coefficients in the Deligne complex

A cocycle α [subscript crossed out or overwritten: illegible mark?]



Uncertain reading. illegible mark

Original source details: GELFAND-P001380

$\in Z^n(X, \mathbb{Z}(i)_{\mathcal{D}})$ consists of a collection

$$\begin{cases} \alpha_0 \in Z^n(X, \mathbb{Z}), \\ \alpha_1 \in C^{n-1}(X, \mathcal{O}), \\ \alpha_2 \in C^{n-2}(X, \Omega^1), \\ \vdots \\ \alpha_i \in C^{n-i}(X, \Omega^{i-1}), \end{cases} \tag{15}$$

compatible in the sense that

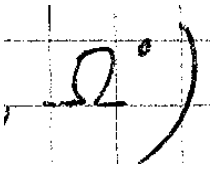
$$\begin{cases} \alpha_0 = \delta\alpha_1, \\ d\alpha_1 = \delta\alpha_2, \\ \vdots \\ \delta\alpha_i = d\alpha_{i-1} \end{cases} \tag{16}$$

Hence

$$d\alpha_i \in Z^{n-i}(X, \Omega^{i,\mathrm{cl}}) \qquad \text{(with values in closed forms)} \tag{17}$$

We can therefore consider

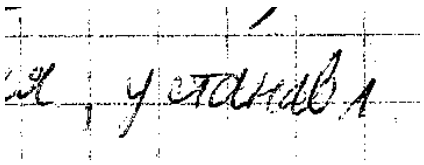
$$\alpha_0 - d\alpha_i \in Z^n(X, \Omega^\circ) \tag{18}$$



Uncertain reading. A small circle is visible above Ω.

Original source details: GELFAND-P001380

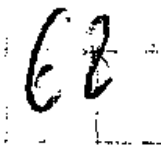
(The rest is a homotopy,



Uncertain reading. «устанавл», short for “establishing,” at the end of the line

Original source details: GELFAND-P001380

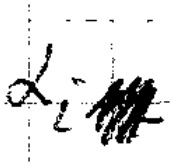
GELFAND-P001381



Uncertain reading. The number at the upper right reads 68 or 62.

Original source details: GELFAND-P001381

$\alpha_1 + \alpha_2 + \ldots + \alpha_i$ [crossed out: illegible mark$^?$]

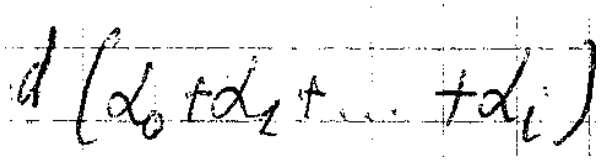


Uncertain reading. illegible mark

Original source details: GELFAND-P001381

$\in C$.

$$d(\alpha_0 + \alpha_1 + \ldots + \alpha_i) = \alpha_0 - d\alpha_i \tag{19}$$



Uncertain reading. In the parentheses in (19) the first term is α_0. The sum in the line above begins with α_1.

Original source details: GELFAND-P001381

So far these are general facts for $\forall$ double complexes.

(see also work by Gabrielov, Gelfand and Losik)

Let us describe the group $H^{2i}(X, \mathbb{Z}(i)_{\mathcal{D}})$ starting from (14).

$$\begin{aligned} 0 \longrightarrow \underbrace{\frac{H^{2i-1}(X,\mathbb{C})}{H^{2i-1}(X,\mathbb{Z}) + F^i}}_{\| \; J^i} &\longrightarrow H^{2i}(X, \mathbb{Z}(i)_{\mathcal{D}}) \\ \longrightarrow \underbrace{H^{2i}(X,\mathbb{Z}) \cap F^i H^{2i}(X,\mathbb{C})}_{\| \; \mathrm{Hg}^i(X,\mathbb{Z})} &\longrightarrow 0 \end{aligned} \tag{20}$$

[Below J^i:] (a complex torus called the ith Griffiths Jacobian)

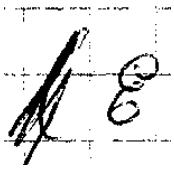
$$H^{2i-1} = F^i \oplus \overline{F}^i$$

$\Rightarrow$ hence the complex structure on J^i.

(21) Example.

$$c_1(\mathcal{E}) = c_1(\det \mathcal{E})$$

g_{ij} — transition functions of the bundle [crossed out: illegible mark?]



Uncertain reading. illegible mark

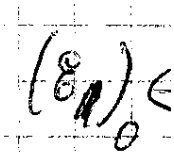
Original source details: GELFAND-P001381

$\mathcal{E}$

(with values in $\mathbb{C}^*$).

$$\begin{aligned} c_1(\mathcal{E}) &\in H^2(X, \mathbb{Z}(1)_{\mathcal{D}}) \\ c_1(\mathcal{E})_0 &\in Z^2(X, \mathbb{Z}) \end{aligned} \tag{22}$$

[In (22), the subscript-like mark before the parenthesis after $\mathcal{E}$: illegible mark?.]



Uncertain reading. illegible mark

Original source details: GELFAND-P001381

$$c_1(\mathcal{E})_1 \in C^1(X, \mathcal{O}), \tag{23}$$

and

$$\delta[c_1(\mathcal{E})_1] = c_1(\mathcal{E})_0 \tag{23'}$$

$$c_1(\mathcal{E})_1 = \log(\deg g_{ij}) \tag{24}$$

— any determination of log.



Original source details: GELFAND-P001381

Uncertain reading. The last letter of the operator in (24) resembles g. The earlier formula (21) uses det.

($c_1(\mathcal{E})_0$ — is found from $(23')$ [crossed out: the mark resembles an overwritten parenthesis?]



Original source details: GELFAND-P001381

Uncertain reading. the mark resembles an overwritten parenthesis

and gives the discrepancy).

This gives an isomorphism

$$c_1 : \mathrm{Pic}(X) \xrightarrow{\sim} H^2(X, \mathbb{Z}(1)_{\mathcal{D}}) \tag{25}$$

GELFAND-P001382

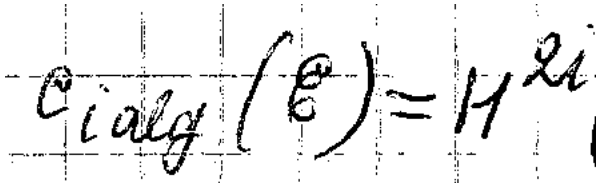
Construction of c_2. (this is already related to the dilogarithm)

c_1 was constructed in 2 steps

$$\mathcal{E} \overset{\det}{\longmapsto} \underset{\text{alg}}{c_1(\mathcal{E})} \in H^1(X, \mathcal{O}^*) \overset{\text{“log”}}{\longrightarrow} H^2(X, \mathbb{Z}(1)_{\mathcal{D}}) \tag{26}$$

In principle it should be

$$\begin{array}{ccc} c_i : \mathcal{E} \longmapsto c_{i\,\text{alg}}(\mathcal{E}) & = & H^{2i}(X, \mathbb{Z}(i)_M) \\ \| & & \\ c_{iM}(\mathcal{E}) & & \\ \text{(motivic)} & & \\ \downarrow \mathcal{L}i_i & & \\ \uparrow & & H^{2i}(X, \mathbb{Z}(i)_{\mathcal{D}}) \\ \text{polylogarithm} & & \end{array} \tag{27}$$

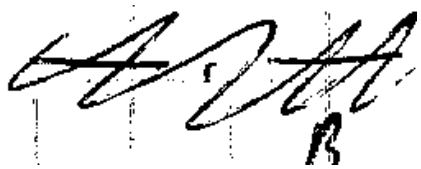


Original source details: GELFAND-P001382

Uncertain reading. In (27) an equals sign appears between $c_{i\,\text{alg}}(\mathcal{E})$ and H^{2i}.

This theory exists up to torsion.

[crossed out: illegible?.]



Original source details: GELFAND-P001382

Uncertain reading. illegible

in it $\mathbb{Z}(1)_M = \mathcal{O}^*[-1]$.

Some requirements

(28) a) $\mathbb{Z}(\text{negative number})_M = 0$

b) $\mathbb{Z}(0)_M = \mathbb{Z}$

c) $\mathbb{Z}(1)_M = \mathcal{O}^*[-1]$

d)

$$\mathbb{Z}(2)_M = \left(\mathbb{Z}(2)^1_M \longrightarrow \underset{\|}{\mathbb{Z}(2)^2_M} \right. \qquad \text{(up to 2-torsion)}$$
$$\wedge^2(\mathcal{O}^*)$$



Original source details: GELFAND-P001382

Uncertain reading. The outer parenthesis after the second term is left open.

$$i \geq 1: \quad H^j(\mathbb{Z}(i)_M) = 0 \qquad \text{when } j \notin [1, i] \tag{29}$$

$$H^i(\mathbb{Z}(i)_M) = \mathcal{K}_i^{\text{Milnor}} \qquad \text{(sheaf of Milnor } K\text{-functors)} \tag{30}$$

K-functor of Milnor.

For a smooth (algebraic) manifold these are sheaves.

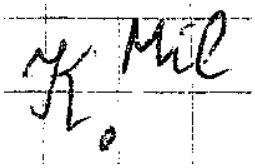
(31) a) $\mathcal{K}_0^{\text{Mil}} = \mathbb{Z}$,

b) $\mathcal{K}_1^{\text{Mil}} = \mathcal{O}^*$

c) [At left: (Steinberg relation).] let $t \in \mathcal{O}^*$ and $1 - t \in \mathcal{O}^*$, then

$$\underbrace{\{t, 1-t\}}_{\substack{\text{Steinberg symbol} \\ \swarrow \\ \text{product in the Milnor algebra}}} = 0$$

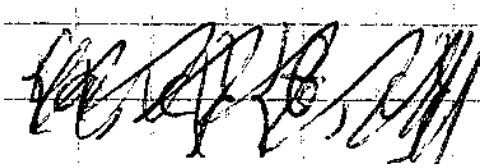
$\mathcal{K}_\circ^{\text{Mil}}$ = free graded algebra with generators [crossed out: $\mathcal{O}^*$] $\mathcal{O}^* = \mathcal{K}_1^{\text{Mil}}$ and the Steinberg relations.



Original source details: GELFAND-P001382

Uncertain reading. In the line on the graded algebra, $\mathcal{K}^{\text{Mil}}$ has a small circle below it that resembles 0.

$\{ab, c\} =$ [crossed out: There appear to be two groups, $\{a, c\}$ and $\{b, c\}$. The sign between them is uncertain.?]



Uncertain reading. There appear to be two groups, $\{a, c\}$ and $\{b, c\}$. The sign between them is uncertain.

Original source details: GELFAND-P001382

?.

[At right:] $\{\ \}$ — addition

GELFAND-P001383

[source number: 69]

(32) The operation $\{\ \}$ is supercommutative in $\mathcal{K}^{\text{Mil}}$.

In (28) d)

[Crossed-out number in the left margin: (33)?.]



Uncertain reading. (33)?

Original source details: GELFAND-P001383

[Entire line crossed out: $\mathbb{Z}(2)^1_M = \mathbb{Z}[\mathbb{P}^1(X)]$.]

$$\mathbb{Z}(2)^1_M = \mathbb{Z}\big[(\mathbb{P}^1 \setminus \{0, 1, \infty\})(X)\big] \,/\, (\text{relations}) \tag{33}$$

[From $\mathbb{P}^1$ a line runs down to the note:] sheaf of functions on X, avoiding the values $0, 1, \infty$.

Differential:

$$\begin{aligned} \mathbb{Z}(2)^1_M &\longrightarrow \mathbb{Z}(2)^2_M \\ t &\longmapsto t \wedge (1 - t) \end{aligned} \tag{34}$$

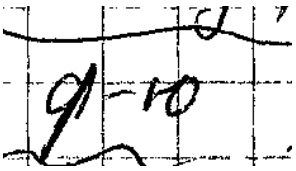
The relation comes from the observation that it lies in the kernel of the Steinberg symbol.

(5-term relation for the dilogarithm)

Let f, g — be 2 functions, nowhere equal to one another and avoiding the values $0, 1, \infty$.

$$(f, g, 0, 1, \infty) \tag{35}$$

(36) 5 distinct points on $\mathbb{P}^1$, omitting one at a time gives 5 cross-ratios, and one has to take their alternating function?.



Uncertain reading. function

Original source details: GELFAND-P001383

This is the 5-term relation.

Instead one can take 5 independent functions.

(37) Assertion. Relation (36) lies in the kernel of the differential.

First devised by Spencer, Bloch.



Original source details: GELFAND-P001383

Uncertain reading. The name before Bloch is written «Спенсер» (Spencer).

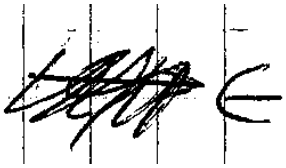
Suslin established the link with K-theory.

The complex is called the Bloch–Suslin complex.

One needs to construct a map

$$\mathcal{E} \longmapsto c_2(\mathcal{E})_M \in H^4(X, \mathbb{Z}(2)_M) \tag{38}$$

[In (38), between $c_2(\mathcal{E})_M$ and $\in$ crossed out: illegible mathematical fragment[?].]



Original source details: GELFAND-P001383

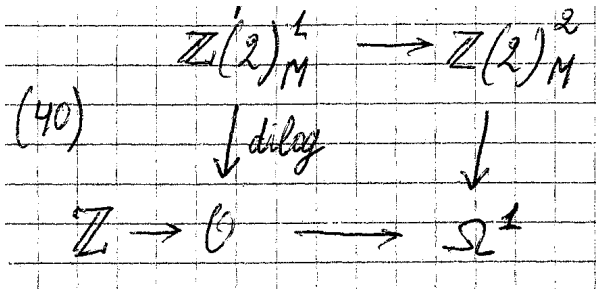
Uncertain reading. illegible mathematical fragment

and a function

$$\mathbb{Z}(2)_M \longrightarrow \mathbb{Z}(2)_{\mathcal{D}} \tag{39}$$

GELFAND-P001384

Let us construct map (39)

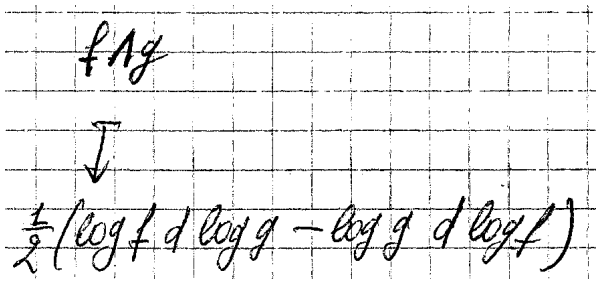


Original source details: GELFAND-P001384

Diagram and typed description.

$$\begin{array}{ccccc} & & \mathbb{Z}(2)^1_M & \longrightarrow & \mathbb{Z}(2)^2_M \\ (40) & & \downarrow{\scriptstyle \text{dilog}} & & \downarrow \\ \mathbb{Z} & \longrightarrow & \mathcal{O} & \longrightarrow & \Omega^1 \end{array}$$

[To the right of the diagram:]



Original source details: GELFAND-P001384

Diagram and typed description.

$$\begin{array}{c} f \wedge g \\ \downarrow \\ \frac{1}{2}(\log f \, d\log g - \log g \, d\log f) \end{array}$$

Chern class.

$$\mathcal{E} \longmapsto c_2(\mathcal{E})_M \in H^4(X, \mathbb{Z}(2)_M), \tag{41}$$

$\dim \mathcal{E} = 2$, $\{U_i\}$, s_i — a section of $\mathcal{E}$ over U_i.

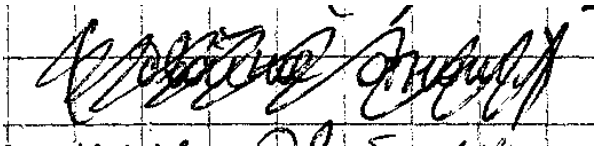
2 components.

$$c_2(\mathcal{E})^2_M \in C^2(\{U_i\}, \mathbb{Z}(2)^2_M)$$
$$c_2(\mathcal{E})^1_M \in C^3(\{U_i\}, \mathbb{Z}(2)^1_M)$$

$(s_1, \ldots, s_4)$ — 4 sections of $\mathcal{E}$.

Consider 4 points $\mathbb{P}(\mathcal{E}) \ni s_1, \ldots, s_4$

and so on. [Crossed out: “cross-ratio”?.]



Original source details: GELFAND-P001384

Uncertain reading. “cross-ratio”

It is constructed using the cross-ratio.